Dataset selection evidence and recommendation

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Status: the student approved the original author dataset on 12 October 2026. This selection evidence is retained alongside Phase1_Strategy_Report.pdf. Preliminary index split candidates below are feasibility evidence; the final plan uses ordered 12/2/2 measured operating blocks with full separator buffers.

Recommendation

Use the original author's Lazzaretti/Costa PV fault data as the primary experimental operating-state dataset, now approved with its limitations documented. Keep the supplied EddieOuyang mirror as verification evidence: both MAT files have identical SHA-256 hashes to the original. Prefer the original source for citation. Its six exported sensor channels support the central ablation question, and labels correspond to physically induced equipment conditions and natural building shade. It is computationally manageable: two files total 28,077,185 bytes. The complete consolidated numeric matrix requires approximately 77 MB at float64; auditing copies and sorting can use more memory.

There is no perfect candidate. The primary recommendation sacrifices exported timestamps and realistic fault prevalence to retain a defensible experimental target. A limited PVDAQ operational slice is a possible secondary *data-quality* study; without verified labels it cannot validate supervised fault recall or F1. Do not merge these datasets. Cross-dataset fault-performance claims require compatible sensors and independently supported labels, neither of which has yet been established for a second dataset.

Evidence categories

Measured findings below come from downloaded files, not advertised sample counts. Documentation claims are marked as such. candidate_audit.json contains the per-variable audit, distribution statistics, class counts, label runs and timestamp summaries. download_manifest.json records actual bytes, URLs, hashes and failed accesses. source_metadata.json records licenses and repository commits. Access date: 12 October 2026.

Candidate comparison

Candidate	Actual files inspected	Sensors/time structure	Labels and main limitation	Decision
PVDAQ/OEDI	Systems metadata 1,862 × 31; bounded operational sample 1 × 15	Electrical/environmental columns; timestamp; site-dependent coverage	Sample has no fault target and four -99999 sentinel cells; one row cannot establish cadence	Optional secondary quality audit; select a bounded site/window later
pvdaq_access	Current README, repository redirect verified	Daily CSV/Parquet data lake access; site IDs required	Access tool, not independent dataset; old API assumed neither needed nor working	Use current NatLabRockies repository if PVDAQ selected
Zenodo 18225566	ZIP 61,486 bytes; scenario CSV 500 × 14; dictionary and supplied script	Normalized scenario indicators; no timestamps or raw electrical channels	Decision/action label; priority is a derived target. Not raw fault-state monitoring	Reject as primary for this research
NASA POWER	Official site/documentation reviewed; no data downloaded	Environmental context, not plant electrical sensors	No verified equipment-fault target	Context only; no fault-label claims
EddieOuyang PV fault mirror	MAT files 1,373,798 × 3 and × 4	Six sensor channels; no exported timestamps	Five operating states; experimentally induced faults and natural shade	Prefer original author's byte-identical source
PV-Fault-DS2 26	Raw CSV 226,175 × 10; labeled CSV 226,175 × 14	Seven acquired channels, two power columns, timestamps with gaps	Two subsystem labels, rule-derived assignment; rare short circuits; power inconsistencies	Alternative if timestamp audit is prioritized, with explicit label caveat
Kaggle Solar Power Generation	Four CSVs in 1,988,756-byte ZIP; dimensions below	Two plants, repeated inverter readings, 15-minute modal cadence and gaps	No fault labels; source license field does not establish a clear reuse license	Not primary supervised target; rights need clarification before redistribution
Zenodo 18326501	One 1,145,740-byte PDF poster only	Poster describes 1-minute electrical and 5-minute IV acquisition	No tabular data file or dataset URL supplied in this record/poster	Not currently verified as usable data

Primary raw audit

dataset_amb.mat is N × 3; dataset_elec.mat is N × 4; N = 1,373,798 in each. Consolidation here means adding columns in the supplied order, not concatenating rows or creating new measurements. Total P = 7 includes the target; predictor P = 6. Rows are repeated measurements from one two-string plant, not 1.37 million independent experiments.

Variable	Meaning and unit	Loaded type	Analytical category and role	NaN cells	Mean	Median	Skewness
vdc1	String 1 voltage, V	float64	Continuous predictor	0	135.1183	37.7763	0.0852
vdc2	String 2 voltage, V	float64	Continuous predictor	0	134.7199	39.8375	0.1158
idc1	String 1 current, A	float64	Continuous predictor	0	2.2769	0.1136	1.0109
idc2	String 2 current, A	float64	Continuous predictor	0	2.0003	0.0441	1.1438
irr	Irradiance, W/m²	float64	Continuous predictor	0	257.6111	1.7364	0.8897
pvt	Reported aggregate module temperature, °C	float64	Continuous predictor	0	22.7728	18.4905	0.5943
f_nv	Operating-state code, unitless	uint8	Nominal categorical target	0	Not meaningful	Not meaningful	Not meaningful

All 1,373,798 rows are complete for exported values, with 0 missing cells out of 9,616,586 cells (0%). There are no exact duplicate consolidated rows and no infinite readings. This establishes neither absence of omitted times nor absence of sensor errors. MCAR/MAR/MNAR cannot be diagnosed from a constant all-zero missingness indicator; the mechanism is unidentifiable from these exported cells.

One irradiance reading is slightly negative (-0.0381 W/m²). It requires a documented near-zero sensor tolerance, not automatic deletion. Negative temperature is physically possible in Curitiba: 24,172 temperature values are below zero, minimum -1.6543 °C. Both irradiance and temperature have 62 zero values; whether these are valid simultaneous readings or sentinels needs further contextual checking. Voltage skewness near zero does not prove a Gaussian distribution: day/night mixtures can be symmetric while highly non-Gaussian. Median imputation is only a provisional fallback; a final choice needs operating-condition analysis, and imputation is not presently necessary for NaNs.

Label	Meaning	Actual count	Percentage
0	Normal	1,162,931	84.6508%
1	Short circuit	5,999	0.4367%
2	Resistive degradation	10,371	0.7549%
3	Open circuit	6,024	0.4385%
4	Building shadowing	188,473	13.7191%

These labels describe experimental operating conditions. Physically induced resistances do not demonstrate years of natural degradation. Shade is an environmental operating condition and should not automatically be counted as a hardware fault. Numeric label codes have no meaningful order.

Timestamp, provenance and splitting risks

No MAT variable contains timestamps, dates or system IDs. The paper describes one-second exported measurements, with electrical RMS aggregation and environmental averaging, but actual time intervals, missing intervals, timezone and drift cannot be independently recovered from these arrays. Preserve acquisition index as metadata. Do not fabricate calendar timestamps or claim verified frequency. Index-based alignment is provisional, supported by the source description, not merely by equal lengths.

The paper's daytime real-data counts total 515,958, whereas the public export totals 1,373,798. Normal and shade counts differ; equipment-fault counts agree exactly. The large low-irradiance fraction is consistent with additional low-light coverage, but the reason for the discrepancy remains unverified. Ask for original timestamps/day boundaries if available; any interpretation must state the discrepancy. The source experiment used scheduled midday induced faults on clear days, so scheduled context and limited weather diversity constrain generalization. Neither Rwanda field reliability nor fault prognosis is demonstrated by this dataset.

A preliminary contiguous 70/15/15 split gives the first partition zero short-circuit examples and only five open-circuit examples. An 80/10/10 ordered split has all classes in each partition:

Provisional partition	Index range, zero-based end-exclusive	N	Normal	Short	Degradation	Open	Shade
Training	[0, 1099038)	1,099,038	922,726	2,458	6,917	2,410	164,527
Validation	[1099038, 1236418)	137,380	124,280	1,180	1,184	1,204	9,532
Test	[1236418, 1373798)	137,380	115,925	2,361	2,270	2,410	14,414

These are feasibility checks, not a finalized evaluation split. Class-preserving proportions are subordinate to acquisition order; do not randomly shuffle adjacent measurements. Day/episode boundaries, purge widths, low-light eligibility and final target require Phase 1 decisions. Label runs can locate boundaries but do not establish independent episodes. All global exploratory findings must be marked descriptive; learned preprocessing and target-based feature selection will use training data only in Phase 2.

PV-Fault-DS226 alternative audit

The actual raw file uses semicolons. It contains timestamp, I1, I2, V1, V2, Irr, T1, T2, P1 and P2. The labeled file adds numeric and text labels for each subsystem; neither counterpart label may be used as a predictor. Timestamps cover 10–23 January 2026 on 12 observed dates; timezone is absent. The median interval is 1.057237 s, maximum 163,354.074604 s. There are no NaNs or duplicate timestamps, but this does not erase large recording gaps. Seven acquired channels are distinct from nine numerical columns: power must not automatically be counted as two additional physical sensors.

Raw power does not equal the supplied voltage-current product within 0.001 W on 25,301 rows for P1 and 19,705 rows for P2. Raw P1 reaches -473.672988 W despite nonnegative exported voltage/current. This is a data-quality concern requiring source clarification, not evidence of a fault by itself.

Class distributions are in candidate_audit.json. Short-circuit labels occur only 53 times in subsystem 1 and 133 in subsystem 2. Preprint labeling uses measured-signal thresholds and fallback rules. Therefore performance against these labels partly measures agreement with the labeling rules, even though the experiment induced physical faults. The preprint is not peer reviewed; descriptions also differ in campaign length (11 versus 12 days). This dataset is more suitable for timestamp/synchronization diagnostics but less suitable for claims of independently verified fault detection.

Other candidate audit details and access actions

Kaggle generation files: plant 1 = 68,778 × 7; plant 2 = 67,698 × 7. Weather files: plant 1 = 3,182 × 6; plant 2 = 3,259 × 6. No exported missing cells occur. Nevertheless, between each entity's first and last timestamp, 15-minute grids have 3,026 and 4,110 absent generation slots, and 82 and 5 absent weather slots respectively. These are missing records, not NaNs. Repeated plant timestamps across inverters are expected; the entity-time key is the relevant duplicate check. Cumulative yields can reveal system identity and long-term information. Zero nighttime power is legitimate. No fault-label field exists. The API reports the license name as “Data files © Original Authors”; this is not an explicit open license grant, so verify intended academic reuse and redistribution with the original source. The archive downloaded without login in this session; no manual login is currently required.

The maintenance subset's 500 rows have scenario/action labels, not timestamps or measured voltage/current. Its priority and severity/status fields can leak decision logic. data_gap_rate is a normalized indicator, not observed per-channel missingness. Zenodo metadata identifies CC BY 4.0.

PVDAQ metadata and the tiny sample verify anonymous HTTPS data-lake access. The 1-row operational sample is a schema and sentinel check only; no representative N, class distribution, cadence or missingness claim is made for a full site. Current access software is at NatLabRockies/pvdaq_access following the NREL URL redirect. Download a bounded site/window after checking file sizes; do not download the whole lake.

Zenodo 18326501 provides a poster without a tabular download link. Obtaining usable data requires a separate authoritative release or asking the listed author for access. No contact has been sent. The MDPI PDF endpoint returned HTTP 403; the original paper was retrieved as full-text XML via Europe PMC. NASA POWER has not been downloaded because it is contextual, not a competing fault-data source.

Approved choice and final Phase 1 plan

The original author dataset is approved **with missing timestamps, count discrepancy and experimental sampling explicitly acknowledged**. The final target is five-class operating-state classification with separate equipment-alert analysis. The final plan allocates 16 measured index operating blocks 12/2/2 and withholds two whole low-irradiance separators; these are not verified dates. See Phase1_Strategy_Report.pdf for exact bounds, class coverage, preprocessing and leakage design. No production pipeline, model training or Phase 2 deliverable has been created. Git/GitHub submission and separate Phase 2 approval remain outstanding.

Sources

1. Supplied Assignment 2 PDF: pp. 1–10, especially Phase 1 rubric pp. 4–5 and submission rules p. 3.

2. Supplied Assignment 1 proposal: pp. 1–2, including sensor ablation equation and project timeline.

3. Original dataset: https://github.com/clayton-h-costa/pv_fault_dataset and downloaded README/LICENSE.

4. Original paper: https://doi.org/10.3390/s20174688; full XML retrieved via https://www.ebi.ac.uk/europepmc/webservices/rest/PMC7506914/fullTextXML.

5. Candidate mirror: https://github.com/EddieOuyang2004/pv_fault_dataset.

6. PVDAQ: https://data.openei.org/submissions/4568; access tool https://github.com/NatLabRockies/pvdaq_access.

7. Maintenance scenarios: https://doi.org/10.5281/zenodo.18225566.

8. DS226: https://doi.org/10.5281/zenodo.19259504 and https://www.preprints.org/manuscript/202607.2348.

9. Solar generation: https://www.kaggle.com/datasets/anikannal/solar-power-generation-data.

10. Benchmark poster: https://doi.org/10.5281/zenodo.18326501.

11. Environmental context: https://power.larc.nasa.gov/.